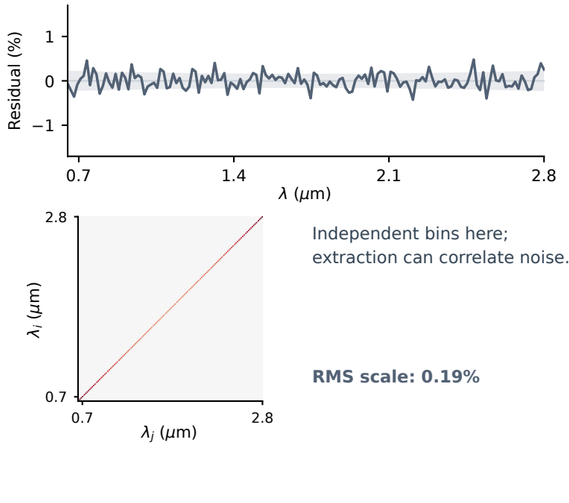
Where does a stellar-spectrum mismatch come from?

A Gaussian error budget: example residuals, wavelength covariance, and sharing across observations

a Measurement noise

EXPOSURE

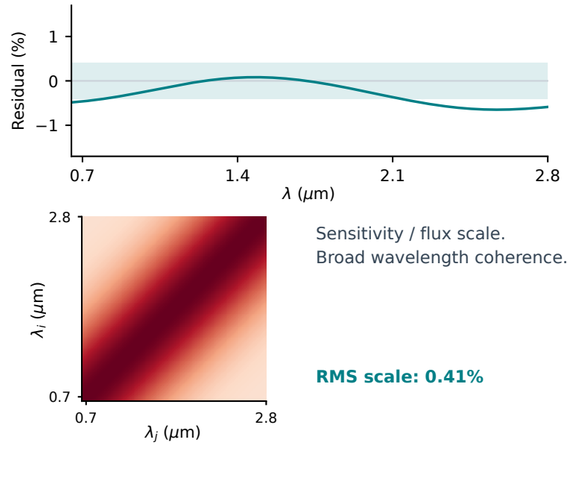


Independent bins here;
extraction can correlate noise.

RMS scale: 0.19%

b Flux calibration

INSTRUMENT / MODE

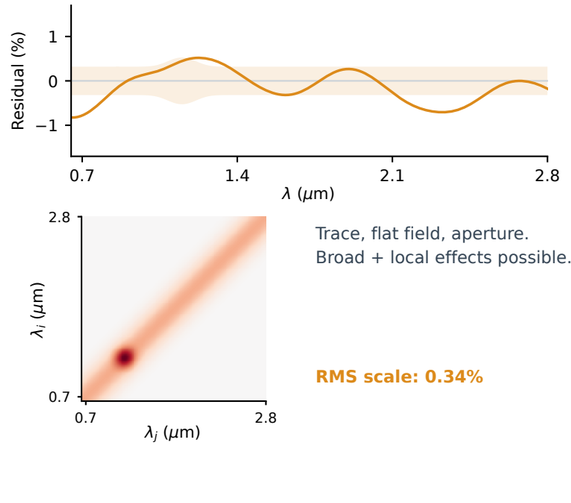


Sensitivity / flux scale.
Broad wavelength coherence.

RMS scale: 0.41%

c Position / extraction

VISIT / DETECTOR

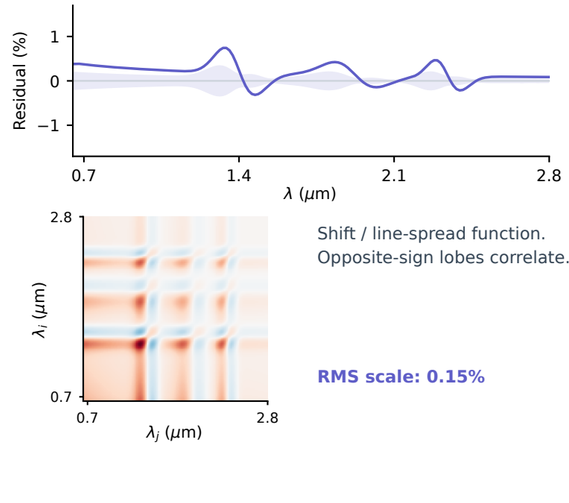


Trace, flat field, aperture.
Broad + local effects possible.

RMS scale: 0.34%

d Wavelength / resolution

EXPOSURE / MODE

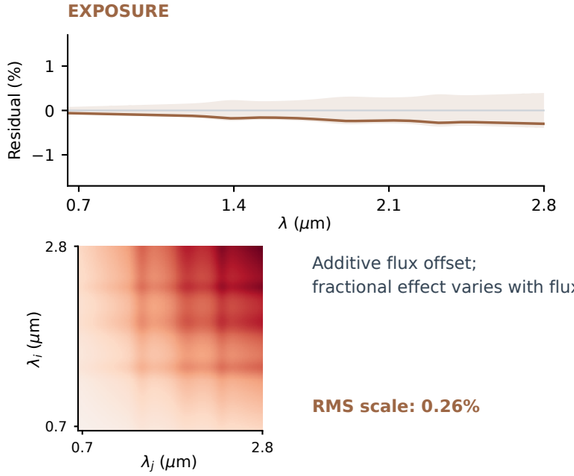


Shift / line-spread function.
Opposite-sign lobes correlate.

RMS scale: 0.15%

e Background residual

EXPOSURE

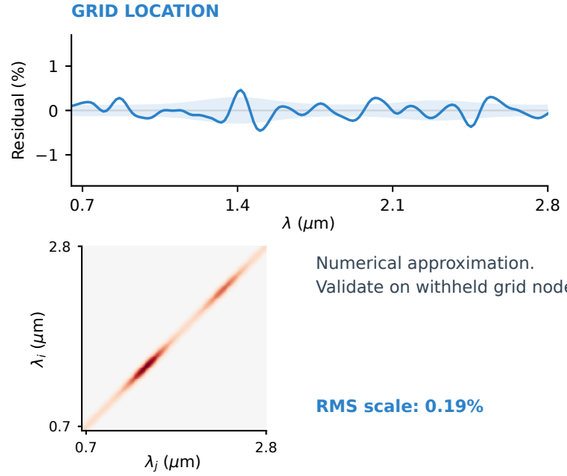


Additive flux offset;
fractional effect varies with flux.

RMS scale: 0.26%

f Grid interpolation

GRID LOCATION

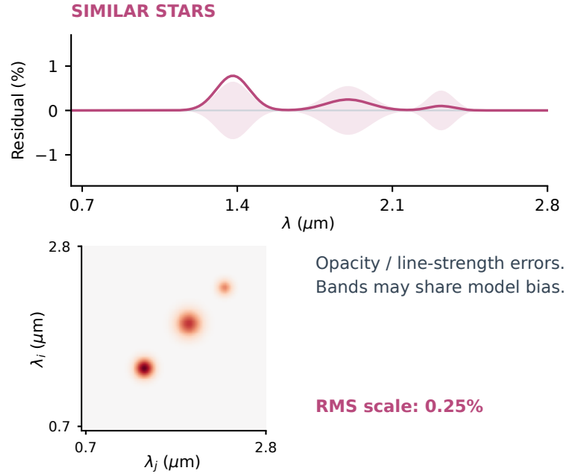


Numerical approximation.
Validate on withheld grid nodes.

RMS scale: 0.19%

g Atmosphere discrepancy

SIMILAR STARS

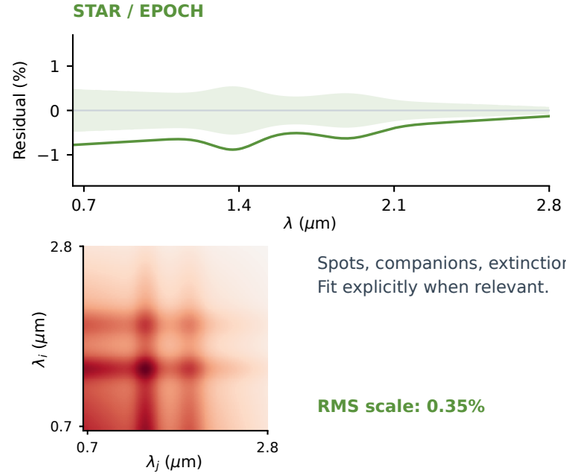


Opacity / line-strength errors.
Bands may share model bias.

RMS scale: 0.25%

h Unmodeled astrophysics

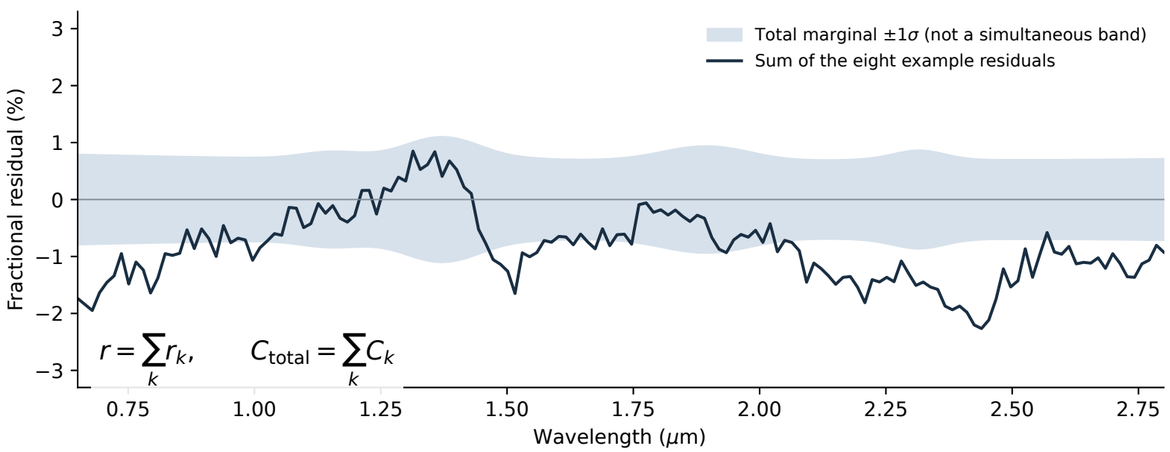
STAR / EPOCH



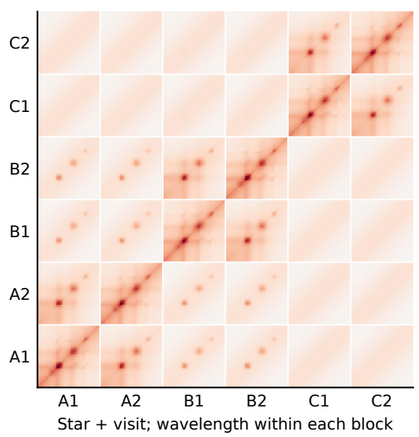
Spots, companions, extinction.
Fit explicitly when relevant.

RMS scale: 0.35%

i Independent latent terms add in covariance



j Sharing across stars / visits



Off-diagonal blocks matter

A1 / A2: repeat visits of star A
B1 / B2: a similar star
C1 / C2: a hotter star

Calibration links every block.
Atmosphere error links similar stars.
Visit errors stay in diagonal blocks.

Shared errors do not average down
as independent measurement noise.

How to read: lines = one Gaussian draw; shading = marginal ±1σ; matrices = covariance shape, each scaled by its own maximum.

Schematic amplitudes, band locations, and sharing assumptions. Error sources are not uniquely identifiable from residual shape alone.

Inspired by the additive-kernel illustration in Rotman et al. (2025), Fig. 2, arXiv:2503.21702. New synthetic stellar-spectrum demonstration.